

# Generation Interconnection System Impact Study Report

## For Queue Project Q21-1963

Roanoke Rapids 69 kV

April 20, 2023

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1963. The Interconnection Customer proposes a 51.6 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 69 kV substation in the North Carolina area. The request was received on December 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 611 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The study did not identify any facilities overloaded as a result of the project. Network upgrade cost: \$0.  
Estimated network upgrade cost: \$0

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

No overloads identified.

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

### **4. Required Network Upgrades and Cost Estimates**

None. Total network upgrade cost allocated to this project: \$0.

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.